

Assignment - CO1-AT-1

Short answer Questions.

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COURSE NAME : fundamentals of data science

1 Data Science and Big data

* A hospital has patient records for many years. how can data Science be applied?

=> Data Science can analyze patient records to identify disease patterns, predict health risks, and improve diagnosis and treatment.

* A bank wants to detect suspicious transactions. how can data Science be applied?

=> Data Science can analyze transaction patterns and detect unusual activities helping banks identify and prevent fraud.

* A school wants to improve student performance. Mention one benefit of data Science in this case.

=> Data Science can identify students who are struggling and help teachers provide personalized support to improve their performance.

* A transport department wants to reduce traffic congestion. how can data Science be applied?

=> Data Science can analyze traffic and GPS data to identify congestion patterns and optimize traffic signals and routes.

* A company has data in Excel tables, customer reviews, images and videos, which facet of data is shown here?

=> this shows variety one of the characteristics of big data because the data exists in different formats.

* A university stores student details in rows and columns
what type of data structure is this?

=> it is structured data, because the information is organized in rows and columns in a fixed format

* A company collects a very large amount of daily user activity data from its mobile app. which big data characteristic is applied?

=> the characteristic is volume, which refers to the huge amount of data being generated and collected.

* A business wants to convert to process large datasets using parallel computation. which big data tool can be used?

=> Value is important because it focuses on extracting useful information and insights from data for better business decision

* A company wants to process large datasets using parallel computation. which big data tool can be used?

=> Apache Hadoop can be used to process large datasets with distributed and parallel computing across multiple computers

* A company wants to visualize big data results for decision-makers. which tool or method can be applied?

=> data visualization tools such as Tableau or Power BI can be used to present big data results through charts, graphs, and dashboards.